Venkata Sai

Software Performance Engineer

venkatasai.jobs@gmail.com | +1(512)-848-0295 | Austin, TX

Professional Experience

Senior Software Engineer *Jul 2025 – Present*

Persistent Systems | High-Performance Runtime Systems

- Designed and implemented core **C++** components for a **high-performance runtime system**, optimizing **application workloads** by 15% through **advanced optimization techniques** for critical operations.
- Developed custom **Python** scripts for **workload analysis** and **performance profiling** of system components, identifying bottlenecks and enabling a 20% reduction in execution time.
- Architected and integrated a new **memory management** module using **C** for a **device driver**, enhancing **system-level programming** efficiency and reducing memory footprint by 10MB.
- Led efforts to **debug** and **optimize** existing **device drivers** using **GDB** and **Valgrind**, improving interrupt latency by 30% for improved **hardware potential**.
- Implemented **compiler optimizations** within the **GCC toolchain** for specific **CPU instruction sets**, yielding a 5% improvement in **code generation** speed and overall application performance.
- Collaborated with **cross-functional teams** to define **performance enhancement algorithms** and **benchmarking** strategies, achieving a 25% improvement in application throughput on new **computer architecture**.
- Developed an **Assembly** (x86) module for critical **low-level operations** in a real-time **Operating System** kernel, improving context switching overhead by 1.2 microseconds.

Software Engineer *Jul 2023 – Jun 2025*

Nagarro | System-Level Performance Optimization

- Developed **C/C++ code analysis** tools to identify **performance improvement opportunities** in critical **software frameworks**, reducing redundant computations and improving execution speed by 18%.
- Implemented **data structures & algorithms** like **hash tables** and **trees** in **Python** for efficient data processing, accelerating **workload analysis** reporting by 40% for engineering teams.
- Contributed to **device driver** development using **C** for embedded **platform level features**, ensuring stable **system calls** and reducing hardware initialization time by 500ms.
- Applied **debugging and profiling techniques** with **GDB** to resolve complex **memory management** issues in **C++** applications, preventing crashes and improving stability by 99.5%.
- Designed and **benchmarked optimization solutions** for **application behavior** using **C++**, integrating **virtualization technologies** to achieve a 10% reduction in VM startup time.
- Wrote **Python** scripts for automating **test and validation** of **performance enhancement algorithms**, streamlining testing pipelines and accelerating release cycles by 2 days.
- Analyzed **CPU instruction sets** and **memory hierarchies** to tailor **optimization techniques** for specific **computer architectures**, boosting data processing throughput by 15% for enterprise clients.
- Integrated **Git** into the development workflow for **multi-developer environments**, ensuring robust **version control systems** and reducing code merge conflicts by 75% across projects.

Software Engineer *Jul 2021 – Aug 2022*

Globant | Application Framework Development

- Developed and maintained **software frameworks** using **Python** and **C++** for various client applications, improving module reusability and accelerating feature development by 20%.
- Assisted in **debugging** and **optimizing application behavior** through **code analysis**, identifying inefficient loops and reducing processing time for critical reports by 15%.
- Implemented new features in **C/C++** applications, focusing on **data structures** and **algorithms** like **graphs**, to improve data retrieval efficiency for client dashboards by 25%.
- Utilized **Git** for **version control systems** in **multi-developer environments**, contributing to **code quality** and ensuring seamless collaboration across engineering teams for timely project delivery.
- Wrote **Python** scripts for **tooling and scripting** to automate routine deployment tasks, reducing manual effort by 30% and improving overall development pipeline efficiency.
- Developed lightweight **C++** components for existing **software frameworks**, focusing on improving **application efficiency** and integration with higher-level systems, reducing processing overhead by 8% for critical functions.
- Collaborated on early-stage **code analysis** initiatives to pinpoint **performance improvement opportunities** within client applications, utilizing basic profiling to enhance overall **application behavior** and responsiveness by 10%.

Skills

Languages: C++, C, Python, Assembly
Frameworks & Tools: Git, GDB, Valgrind, GCC Toolchain
Concepts: System-Level Programming, Memory Management, Compiler Technologies, Performance Profiling, Computer Architecture, Data Structures & Algorithms, Operating Systems, Optimization Techniques, Virtualization

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India